Generative AI for OEMs: Revving up your Dream car

TEAM ASCA

INTRODUCTION

In this competitive market, product personalization is essential as customer expectations continue to rise. Our idea focuses on this critical business need, addressing two key aspects:

1. Enhancing Customer Satisfaction: We empower customers to personalize products, delivering a unique shopping experience that aligns with their preferences.

2. Improving Manufacturing Efficiency: Our AI system streamlines product customization, reducing costs, and enhancing production efficiency, especially for Original Equipment Manufacturers (OEMs).

IDEA DETAILS

IDEA:

We are creating an all-inclusive automobile personalization assistant that uses natural conversations to collect user preferences, suggests configurations, and converts them into requirements for automated implementation. This innovative approach simplifies preference consolidation, replacing manual processes with a 24/7 virtual personalization expert. Users also receive a visual preview of their customized product before purchase, ensuring satisfaction.

CONTEMPORARY (SIMILAR) SOLUTIONS:

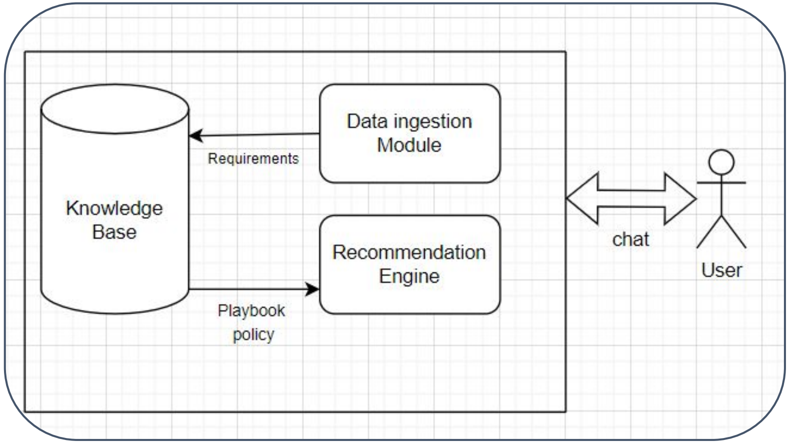
1. https://www.bentleymotors.com/en/misc/car-configurator.html
2. https://www.porsche.com/international/modelstart/
3. https://www.landrover.in/build-your-own/index.html
4. https://carconfigurator.ferrari.com/en_EN

SOLUTION OVERVIEW

MODULE DESCRIPTION

An intelligent form is deployed and used to get an idea about the personal tastes of the user following which suggestion prompts and follow ups are provided. Based on this interaction customers will be able to visualize their wants and needs and try out a plethora of different options adding another dimension of user experience and luxury.

ARCHITECTURE



TECH STACK

SOFTWARE REQUIREMENTS:

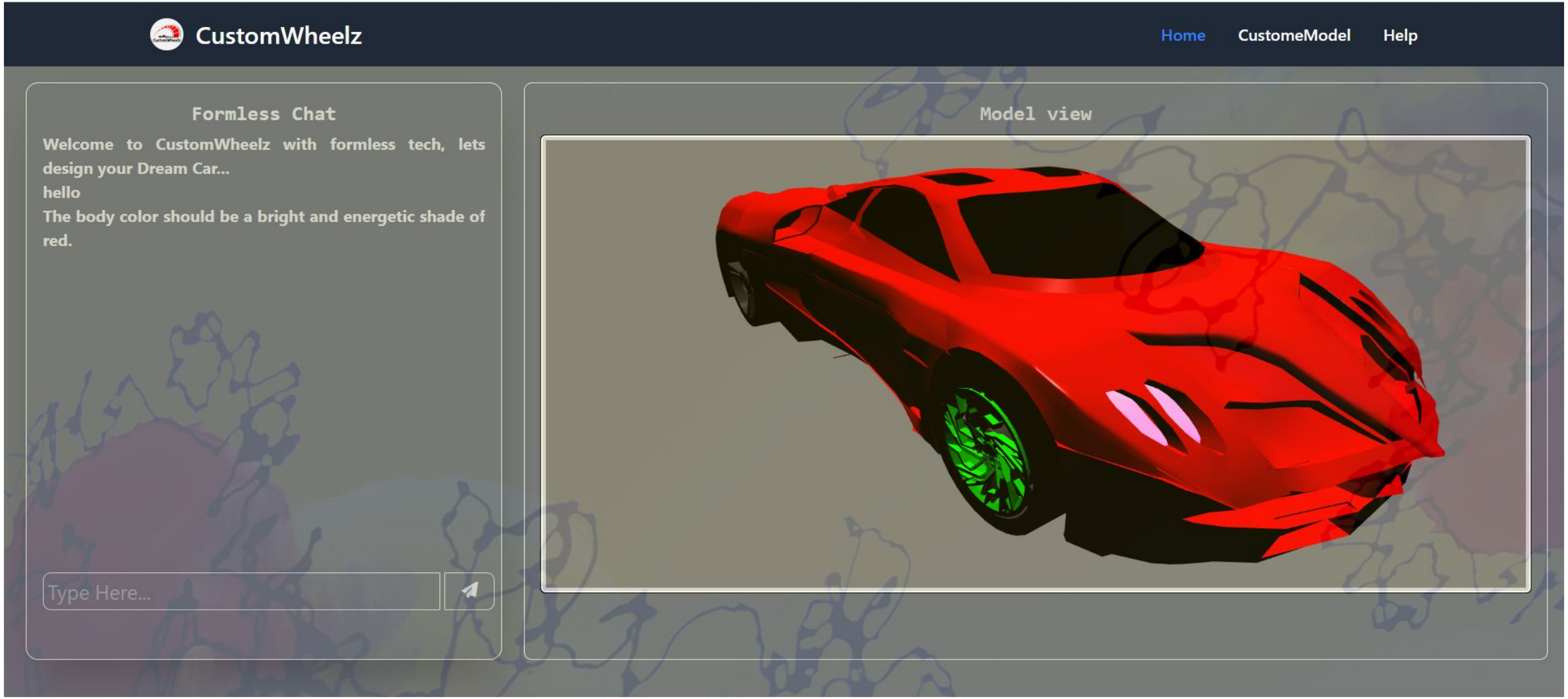
- Django
- LANGCHAIN
- OPEN-SOURCE LLMS(LLAMA-2)
- BABYLON.JS
- UNREAL ENGINE
- PYTHON
- Spacy

LIMITATIONS

Current prototype has following limitations:

1. <u>Single Car Model</u>: The prototype is based on a single car model because readily available 3D object files were limited, restricting the range of vehicle options that users can personalize.
2. <u>Limited Customization Options</u>: The prototype offers only basic customization options, such as color and skin, due to constraints related to the availability of 3D object files. This restricts the extent to which users can personalize their vehicles.
3. <u>Lack of Personalization Features</u>: The current prototype lacks the ability to incorporate playbooks and provide tailor-made, natural chat interactions with users. This limitation hinders the level of personalization and engagement that can be achieved.

DEMO


CustomWheelz
Home
CustomeModel
Help
Formless Chat
Welcome to CustomWheelz with formless tech, lets design your Dream Car...
hello
The body color should be a bright and energetic shade of red.
Type Here...
Model view

FUTURE ENHANCEMENTS

1. <u>Improved Customization Options:</u>

   - Expand the product to accommodate a wider range of vehicle models.

   - Offer advanced and wider customization features, including interior details, accessories and integrating advanced AI like styleGAN for more personalised visual preview.

2. <u>Enhanced Personalization Experience:</u>

   - Implement playbooks and natural chat interactions according to company’s policy.

   - Integrate car configuration APIs for real specifications and pricing data.

CONCLUSION

As the market shifts toward greater product personalization, our idea is positioned to address this crucial business need effectively.

We offer a win-win solution:

1. Empowering customers to personalize products, enhancing their satisfaction and loyalty.
2. Streamlining manufacturing processes with AI, reducing costs, and improving efficiency, especially for OEMs.
3. By bridging the gap between customer expectations and operational efficiency, our concept ensures that businesses stay competitive in a rapidly evolving market.

THANK YOU